Sankaran Vaidyanathan

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Research Interests

Developing causal inference methods, particularly at the instance-specific and path-specific level, for mechanistic interpretability and evaluation of complex AI systems such as LLMs and RL agents.

Education

University of Massachusetts Amherst *Sept 2021–May 2027*
Ph.D. in Computer Science

University of Massachusetts Amherst *Sept 2019–May 2021*
M.S. in Computer Science

Anna University *Aug 2013–May 2017*
B.E. in Electrical and Electronics Engineering

Experience

Research Intern *Cambridge, MA*
Basis Research Institute *June 2026–present*

Research Fellow *Remote*
AI Safety Camp *Feb 2025–Aug 2025*

Research Assistant *Amherst, MA*
Knowledge Discovery Lab & Safe, Confident, and Aligned Learning + Robotics Lab *May 2020–present*
University of Massachusetts Amherst

Project Associate *Chennai, India*
Robert Bosch Center for Data Science and AI *July 2017–June 2019*
Indian Institute of Technology Madras

Selected Publications

The Curse of Multiple Mediators: Hidden Interaction Effects in Activation Patching
Sankaran Vaidyanathan, *David Arbour*, *Aaron Mueller*, *Scott Niekum*, *David Jensen*
arXiv:2606.27510 (under review)

Hierarchical Experimentalist Agents
Abhranil Chandra, ***Sankaran Vaidyanathan***, *Utsav Dhanuka*, *Varun Gandhi*, *Scott Niekum*
RLC Workshop on Reinforcement Learning Beyond Rewards: Towards Scalable General-Purpose Agents, 2026

Detecting and Characterizing Planning in Language Models
Jatin Nainani, ***Sankaran Vaidyanathan***, *Connor Watts*, *Andre N. Assis*, *Alice Rigg*
NeurIPS Workshop on Mechanistic Interpretability, 2025

Judging the Judges: Evaluating Alignment and Vulnerabilities in LLMs-as-Judges
Aman Singh Thakur, Kartik Choudhary*, Venkat Srinik Ramayapally**, ***Sankaran Vaidyanathan***, *Dieuwke Hupkes*
ACL Workshop on Natural Language Generation, Evaluation, and Metrics (GEM), 2025

Quantitative LLM Judges
Aishwarya Sahoo, Jeevana Kruthi Karnuthala*, Tushar Parmanand Budhwani*, Pranchal Agarwal**, ***Sankaran Vaidyanathan***, *Alexa Siu*, *Franck Dernoncourt*, *Jennifer Healey*, *Nedim Lipka*, *Ryan Rossi*, *Uttaran Bhattacharya*, *Branislav Kveton*
arXiv:2506.02945 (under review)

Automated Discovery of Functional Actual Causes in Complex Environments
*Caleb Chuck**, ***Sankaran Vaidyanathan****, *Stephen Giguere*, *Amy Zhang*, *David Jensen*, *Scott Niekum*
arXiv:2404.10883 (under review)

Collaborative Research Projects

Probabilistic Actual Causality in Transformers *June 2026–present*
Basis Research Institute

- Identifying redundant mechanisms in LLMs using actual causality and probabilistic programming.

Interphyre *Sep 2025–June 2026*
Safe, Confident, and Aligned Learning + Robotics Lab, UMass Amherst

- Built a 2D physics puzzle environment for interpretability and agentic experimentation research, with levels as Python code, full simulation state access, and an intervention API for paired counterfactual trajectories.

Evaluation Awareness in LLM Safety Benchmarks *Feb 2026–May 2026*
Microsoft

- Showed that LLMs internally represent whether or not they are being tested even when they fail to verbalize it, and that models better at detecting tests are harder to manipulate into producing harmful outputs.

Quantitative LLM Judges *Feb 2025–May 2025*
Adobe Research

- Developed a lightweight post-hoc framework that improves LLM-as-a-Judge alignment with human scores using generalized linear models on judge embeddings, avoiding expensive fine-tuning.

Evaluating Alignment and Vulnerabilities in LLMs-as-Judges *Feb 2024–Dec 2024*
Meta

- Demonstrated that simple lexical metrics and smaller models can match large LLM judges at ranking model outputs, and that standard agreement metrics mask systematic misalignment with human scores.

Analysis and Prediction of Cognitive Load During Cardiac Surgery *May 2023–May 2024*
National Institute of Health and Harvard Medical School

- Predicted cognitive load in surgical teams from time-series heart rate variability data, and identified key predictive features using SHAP and feature ablation validated against clinical expertise.

Teaching Experience

- Guest Lecture: Mechanistic Interpretability, *COMPSCI 690S: AI Alignment* *Spring 2026*
- Guest Lecture: MCMC and Hamiltonian Monte Carlo, *MATH 605: Probability Theory* *Fall 2021*
- Teaching Assistant: *Artificial Intelligence, Probabilistic Graphical Models, Data Structures, Decarbonization and Data Science*

Service and Outreach

- Reviewer: NeurIPS 2026, ICML 2026, AAAI 2026, NeurIPS 2025 Mechanistic Interpretability Workshop
- Founder, UMass Mechanistic Interpretability Reading Group *2025–2026*
- Co-organizer, UMass Machine Learning and Friends Lunch *2019–2020, 2023–2026*
- Mentor, UMass Ph.D. Applicant Support Program *2021–2025*
- M.S. Graduate Representative, UMass College of Information and Computer Sciences *2020*
- Volunteer Pen-Pal, Letters to a Pre-Scientist *2024–2026*

Technical Skills

Languages and Frameworks: Python, C++, PyTorch, Pyro-PPL, ChiRho, Box2D, NNsight

Design: Figma, Inkscape, Affinity Designer, Adobe After Effects

Relevant Graduate Coursework

Causal Inference, Bayesian Statistics, Probabilistic Graphical Models, Reinforcement Learning, Advanced Natural Language Processing, Neural Networks: A Modern Introduction, Machine Learning, Artificial Intelligence, Optimization in Computer Science, Probability Theory, Math Statistics, Research Methods in Empirical Computer Science, Distributed and Operating Systems, Quantum Information Systems, Fixing Social Media